MD RAKIBUL HASSAN

Software Engineer | Python, Data Processing & Automation
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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Software Engineer and M.Sc. student in High Integrity Systems with professional experience in Python development, data processing, automation, and backend-integrated applications. Experienced in developing data-driven software, REST APIs, and containerized Python applications using Docker, Linux, Git, and CI/CD pipelines. Strong analytical and problem-solving skills, with a focus on applying machine learning, real-time data processing, and reliable software engineering to automotive research and intelligent vehicle technologies.

TECHNICAL SKILLS

- **Programming & Frameworks:** Python, Dart, JavaScript, SQL, C++, Flutter
- **Machine Learning & AI:** NLP, Scikit-learn, XGBoost
- **Data Engineering & Automation:** Pandas, NumPy, Data Visualization**,** Web Scraping, BeautifulSoup, Playwright, Data Processing, Data Cleaning, JSON Processing, GPS Data
- **MLOps & Deployment:** Model Evaluation**,** Docker, CI/CD Pipelines, Linux (CLI & Administration), Server Setup & Deployment
- **Backend & APIs:** FastAPI, RESTful APIs, Third-party AI Service Integration
- **Cloud & Infrastructure:** AWS, Azure.
- **Development Tools:** Git/GitHub, GitLab, Postman, SSH, Jira, Trello

PROFESSIONAL EXPERIENCE

Software Developer (Working Student) Frankfurt Am Main, Germany
Enviotech GmbH *July 2026 – Present*

- Developing Python-based data extraction and automation pipelines using BeautifulSoup and Playwright to collect and process data from municipal websites and documents.
- Transforming unstructured data into structured information through data cleaning, filtering, classification, and scoring for an automated lead intelligence and reporting system.
- Generating structured Excel reports and data extracts, including priority classifications, supporting evidence, and run summaries.
- Troubleshooting scraping, data-quality, and reporting issues across heterogeneous websites and document formats; translating business requirements into software solutions with the project lead.

Software Developer Offenbach, Germany
Louco Event Media GmbH *March 2026 – June 2026*

- Developed and maintained production software for Android and iOS, integrating REST APIs and backend services for authentication, notifications, and personalized user experiences.
- Investigated and resolved production issues involving APIs, authentication, notifications, and application performance.
- Collaborated with backend developers in Agile Scrum to design, test, and deliver software, including sprint planning, testing, and production releases.
- Managed Android and iOS deployments using Google Play Console and TestFlight.

HiWi – Mobile Application Developer Mainz, Germany
Johannes Gutenberg University Mainz *September 2025 – February 2026*

- Developed a cloud-connected research platform supporting environmental monitoring and data collection.
- Integrated AWS cloud services, GPS data, and external APIs.
- Built scalable software for collecting, processing, and visualizing environmental information for academic research projects.

Software Developer Kuala Lumpur, Malaysia
Teczo Sdn. Bhd. *February 2022 – August 2022*

- Developed production software applications serving multiple customers, integrating REST APIs and backend services.
- Improved software quality and maintainability through modular architecture and optimized application performance.
- Collaborated with cross-functional teams using Git and Agile Scrum.

EDUCATION

Master of Science in High Integrity Systems Frankfurt, Germany
Frankfurt University of Applied Sciences — Major: Computer Science *Ongoing*

Bachelor of Computer Science Malaysia
International Islamic University Malaysia — Major: Software Engineering *August 2023*

PUBLICATIONS

- BEST4U.COM: A Creative Platform for Secondhand Items Transaction — May 2022
- STEKO-SD: A Mobile Health Solution to Assist Self-Checking of Metabolic Syndrome — Ongoing

PROJECTS

Real-Time Driver Stress & Cognitive Load Monitoring System
Python, Pandas, NumPy, Scikit-learn, FastAPI, Docker, Streamlit

- Project scope: physiological signal analysis, feature extraction, and machine learning-based driver stress classification using public research datasets.
- Architecture: offline model evaluation, simulated streaming inference, and real-time prediction visualization.
- Target deployment: containerized Python inference service with FastAPI and a research monitoring dashboard.

Scalable Data Pipeline for Satellite & GPS Telemetry
Python, AWS, GPS, Cloud

- Developed a cloud-connected environmental monitoring platform integrating GPS and satellite information with AWS cloud storage.
- Designed scalable software supporting real-time processing of environmental and location data.

Smart Infrastructure Lead Intelligence
Python, BeautifulSoup, Playwright, Data Processing, Testing, Git

- Developed a modular Python system for automated collection and processing of structured information from municipal websites and public-sector sources.
- Implemented static and dynamic extraction with BeautifulSoup and Playwright, including crawling, URL normalization, deduplication, retry handling, and rule-based scoring to classify and prioritize results.
- Built automated component tests and applied Git-based version control, logging, and configuration management to improve reliability and maintainability.

LANGUAGES

Bengali (Native) | English (Fluent) | German (Basic)